Final Reflection

Two Azure AI service problems hit while building the boarding kiosk, and how they were resolved

Problem 1: Document Intelligence rejected my boarding-pass label files

What happened. I generated the training labels in code instead of clicking through the labeling site. For each of 12 synthetic boarding passes I ran the layout model, found the known values among the recognized words, and wrote `<pass>.pdf.labels.json`, `<pass>.pdf.ocr.json` and `fields.json` to Blob Storage. Building the custom template model then failed with:

```
ModelBuildError: Can't find any valid labels for provided dataset. Generic error during processing labels for training-pass-01.pdf … fields.json is present but has wrong schema.
```

The message names no field or line. Two attempts at "fixing the obvious thing" changed nothing: wrapping the layout result in the Studio envelope (`status`, `createdDateTime`), and requesting `utf16CodeUnit` offsets. Removing `.ocr.json` altogether gave a different error, `TrainingContentMissing`, which showed the OCR file is mandatory.

How I resolved it. I stopped changing one thing per training run. Instead I uploaded copies of the same six passes under separate blob prefixes, each with a single variation of the label file, and started one build per prefix with the `prefix` parameter of `AzureBlobContentSource`. One variant built successfully: label files **without the top-level `"$schema"` key** (and a `fields.json` with only a `fields` array). With that change the real model, `boarding-pass-v1`, trained with an estimated accuracy of 0.995 on all 10 fields and extracted all six real passes correctly. I added a unit test that asserts the label file has exactly the keys `document` and `labels`, so the regression cannot return.

Lesson. When a service only says "generic error", run controlled variants side by side rather than guessing one change at a time. Each build took about a minute, so the comparison cost minutes, not hours.

Problem 2: Vision services blocked by Azure Policy in the first lab subscription

What happened. I started in the subscription that was already signed in to the Azure CLI. Storage and Document Intelligence deployed fine. Every attempt to create Face, Custom Vision or Computer Vision failed with:

```
RequestDisallowedByPolicy: Resource 'kiosk-face-1aq' was disallowed by policy … "Restrict Vision and Speech AI services"
```

Calling the Face API through a multi-service AI Services account returned `401 Access denied`. Registering the `Microsoft.VideoIndexer` provider failed with `AuthorizationFailed`. A second lab subscription in the same tenant had the same management-group policy. The subscription belonged to a different (generative AI) course, so its policy only allowed that course's services.

How I resolved it. I diagnosed it before writing any face code: I read the policy assignment named in the error and probed each service with one create call. Then I moved only the blocked services to the computer-vision course's own lab subscription (Udacity CloudLabs). There I created Face, Custom Vision (training and prediction) and a Video Indexer account. Video Indexer needs a storage account and a *Storage Blob Data Contributor* role for its managed identity. I also checked that Face *detect* returned a `faceId`, since verify and identify are Limited Access features. Document Intelligence and the project's Blob Storage stayed where they were, so the solution now spans two subscriptions. Each service's endpoint and key come from `.env`, and the teardown script deletes resources in both.

Lesson. In a managed environment, check which services the subscription allows (policy and provider registration) before designing around them. The error message names the exact policy, which settles whether to keep retrying or move.

Other issues worth noting

- **Retired services.** Azure Video Analyzer was retired in 2022. Video Indexer replaced it, and it no longer reports facial emotions. Emotions now come from speech, so the silent identity video produces only *Neutral* sentiment. The feedback label is therefore "neutral".
- **Custom Vision "Failed" iteration.** The first training failed because the images carried the `lighter` tag at image level with no bounding boxes, and object detection needs regions. Re-uploading the images with 60 boxes fixed it. The boxes are kept in `data/lighter_boxes.json`, so the dataset is reproducible.
- **Domain shift in lighter detection.** Training photos are close-ups of single lighters, while the test trays show small lighters among clutter. The quick-trained model had 100% precision but missed three of the five test lighters at the 0.5 cutoff. I kept the threshold fixed rather than lowering it after seeing test results, and retrained with Advanced training (1-hour budget) instead. Recall rose from 17.6% to 41.2% and mAP from 81.6% to 92.4%, precision stayed at 100%, and the model now catches three of five test lighters. Publishing the new iteration first failed with "The name is in use by another published iteration", so the publish step now unpublishes the old iteration first. For a few seconds after the switch, the prediction endpoint still answered some requests from the old iteration, so I re-scored until the results matched the new model's quick test.